

9 **B** Loss in GPE = gain in KE + energy dissipated by frictional forces

$$\text{energy dissipated by frictional forces} = mgh - \frac{1}{2}mv^2$$

$$= (500)(9.81)(30) - \frac{1}{2}(500)(11)^2$$

$$= 1.2 \times 10^5 \text{ J}$$

10 **B** Energy required per second to raise water